TEXORA · LECTURE SHEET

Carpet Testing and Fibre Identification

A carpet is a three-dimensional textile whose value lies in its pile. This sheet covers how pile construction is measured, how thickness, compression and tuft anchorage are tested under load, how appearance retention is simulated, and how the fibres in a carpet are identified by burning, microscopy and solubility.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 8 · LEVEL 3 TERM 2

1. Carpet construction — what is being tested

Most modern carpet is tufted: pile yarn is needled through a primary backing to form loops (loop pile) or cut loops (cut pile), the tufts are locked in with a latex compound, and a secondary backing is laminated on for stability and dimensional control. Woven carpets (Axminster, Wilton) interlace pile and backing in one operation, but the test logic is the same: a working **pile** layer above a **backing system**. Carpet quality is dominated by how much pile there is, how densely it is packed and how firmly it is held — three different quantities that must be measured separately.

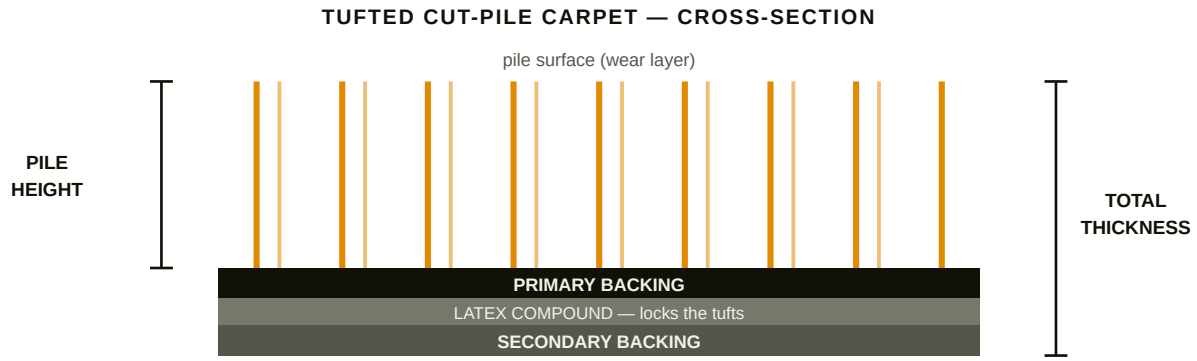


Figure 1 — Cross-section of a tufted cut-pile carpet. Pile height and total thickness are different measurements; pile mass excludes the backing system.

1.1 Pile height, tuft density and pile mass

- **Pile height (pile thickness)**: the length of pile above the backing, measured by a thickness gauge difference method or by extracting and measuring tufts. Longer pile feels luxurious but crushes and

shows shading more readily.

- **Tuft density**: tufts per unit area, obtained from the machine gauge — rows (tufts across the width, per inch) multiplied by stitches per inch along the length. Density, not height alone, decides how firmly the pile supports a foot and resists crushing.
- **Pile mass per unit area**: the mass of pile yarn in 1 m^2, obtained by shearing or chemically removing the pile from a known area and weighing, or by difference between total carpet mass and backing-system mass. Pile mass is the single best proxy for carpet value and wear life — it is the amount of material available to be worn away.

$$\text{Tuft density (tufts/in}^2\text{)} = \text{rows per inch} \times \text{stitches per inch}$$

$$\text{Pile mass per unit area} = \text{total mass per unit area} - \text{backing-system mass per unit area}$$

WORKED EXAMPLE — TUFT DENSITY, PILE MASS AND THICKNESS LOSS

A tufted carpet is made at 8 rows/inch with 9 stitches/inch. Tuft density = 8 × 9 = **72 tufts/in^2** (about 111,600 tufts/m^2, since 1 m^2 = 1,550 in^2).

Total carpet mass = 2,400 g/m^2; backing system (primary + latex + secondary) = 900 g/m^2. Pile mass = 2,400 − 900 = **1,500 g/m^2** — a heavy domestic / light contract pile weight.

Thickness before dynamic loading = 9.6 mm; after loading = 8.1 mm. Loss = 9.6 − 8.1 = 1.5 mm; % loss = 1.5 / 9.6 × 100 = **15.6 %**. A specification limiting loss to 20 % is met — but read together with recovery (Section 3): a carpet that loses 15.6 % under load and recovers most of it overnight performs better than one that loses 10 % permanently.

2. Thickness and thickness loss — ISO 1765 principle

Carpet thickness is measured with a thickness gauge that presses on the pile with a standard presser foot at a standard low pressure (ISO 1765 covers thickness measurement of textile floor coverings). Because pile is compressible, the pressure and foot size are part of the result — thickness quoted without them is incomplete. Two related tests decide serviceability:

- **Thickness under increased load**: thickness is re-measured at higher pressures to build a compression curve. A carpet whose thickness collapses at modest pressure has a low-density pile that will flatten quickly in a doorway.
- **Thickness loss after dynamic loading**: a loaded roller or weighted hammer repeatedly impacts the specimen (simulating foot traffic), and thickness is measured before and after a set number of impacts. The percentage loss ranks resistance to flattening. Dense, resilient pile — wool, or well-twisted nylon at high density — loses least; low-density polypropylene loop pile loses most.

3. Compression and recovery

Compression testing loads the pile progressively and records the load–thickness curve; recovery testing removes the load and follows thickness as it returns over minutes to hours. Three readings describe underfoot quality: **compressibility** (how much the pile yields — too little feels hard, too much feels unstable), **work of compression** (area under the loading curve — the cushioning energy), and **percentage recovery** after a rest period. Wool pile recovers outstandingly because of fibre crimp and resilience; nylon is close behind; polypropylene recovers poorly and flattens permanently in traffic lanes, which is why cheap polypropylene carpets show a worn path even when little pile has actually been lost. Compression–recovery behaviour, not thickness alone, is what the Kawabata-style compression parameters of Topic 9 formalise for flat fabrics.

4. Tuft bind — ISO 4919

Tuft bind (tuft withdrawal force, ISO 4919) is the force needed to pull a single tuft — or one leg of a loop — out of the backing. A tuft is gripped and withdrawn vertically on a tensile tester and the peak force recorded; a set of tufts across the specimen gives a mean and a minimum. Weak tuft bind means the latex has not locked the tufts: the carpet will shed tufts to vacuum cleaners, castor wheels and snagging, loops will pull out in runs, and cut edges will fray. Bind depends on latex quantity and curing, backing penetration and tuft geometry — loop pile normally withdraws at higher force than a single cut tuft leg, so the specification must state which is gripped. In the worked specification of this sheet, tuft-bind readings averaging **29 N** represent firm anchorage for a cut-pile product; the *minimum individual* value matters as much as the mean, because failure starts at the weakest tuft.

5. Appearance retention — hexapod and castor-chair tests

Thickness loss measures flattening; appearance retention judges the whole visual change — flattening, fuzzing, pilling, colour and texture change — after simulated traffic. In the **hexapod tumbler test** specimens line a drum in which a heavy hexagonal tumbler with rubber feet rolls over them for thousands of revolutions. In the **castor-chair test** a loaded chair on standard castors rolls repeatedly over the carpet, reproducing the severe, twisting wear under office chairs — the harshest common service. Tested specimens are graded against photographic or retained standards, commonly on a 5-to-1 scale, and the grade after a stated number of cycles classifies the carpet for end use: heavy contract, general contract, heavy domestic and so on. A carpet can retain thickness yet grade poorly because its texture or colour changed — which is why appearance grading complements, and does not duplicate, the thickness-loss test. Fibre choice dominates retention: wool and nylon retain appearance longest; solution-dyed fibres retain colour best because the pigment is inside the fibre.

6. Fibre identification for carpet fibres

Carpet specifications are written by fibre — wool, nylon, polypropylene, acrylic and their blends price and perform very differently — so the laboratory must be able to verify what a carpet is actually made of. Identification proceeds from quick, cheap, non-destructive tests to definitive chemical tests, always on pile yarn extracted from an inconspicuous place.

6.1 Burning behaviour

A few fibres are held at the edge of a small flame, in the flame, and after removal. Observations — approach behaviour (melting, shrinking, curling), odour, residue — give a first classification. Burning tests are indicative, not proof: blends and finishes distort them, and flame-retardant treatments change behaviour entirely.

6.2 Microscopy

Longitudinal microscopy separates natural from man-made at a glance: wool shows overlapping surface scales; manufactured fibres are smooth rods, often with delustrant particles, and their cross-sections (round, trilobal — common in carpet nylon to hide soil — or hollow) identify the spinning type. Carpet fibres are deliberately engineered in cross-section, so section shape is strong supporting evidence.

6.3 Solubility

Definitive identification uses selective solvents: each fibre dissolves in reagents that leave the others intact, so a blend can be separated quantitatively by successive dissolution and weighing. The scheme is applied in a fixed order, with known controls alongside, because aged, dyed or finished fibres dissolve more slowly than virgin standards.

TEST	WOOL	NYLON (POLYAMIDE)	POLYPROPYLENE	ACRYLIC
Approach to flame	Curls away, slow to ignite	Melts, shrinks away	Melts, shrinks away rapidly	Melts, then burns
In flame / odour	Burns slowly, self-extinguishes; smell of burning hair	Melts, burns slowly; celery-like odour	Burns with melting drips; paraffin-wax odour	Burns rapidly, sputtering; acrid odour
Residue	Crushable black char bead	Hard grey bead, cannot crush	Hard pale bead	Hard irregular black bead

Microscopy	Overlapping scales visible	Smooth rod; often trilobal section in carpet yarn	Smooth rod, round section	Smooth rod, often dog-bone / bean section
Key solubility	Dissolves in hot dilute NaOH / sodium hypochlorite; resists acids	Dissolves in concentrated formic acid / HCl; resists alkali	Resists almost all reagents at room temperature; dissolves only in hot aromatic solvents	Dissolves in dimethylformamide (DMF); resists acids and weak alkali
Carpet character	Best resilience and appearance retention; premium price	Best abrasion resistance; the contract standard	Cheapest, stain-resistant, poor recovery — flattens	Wool-like handle, good colour; moderate wear

The practical scheme is: *burn to sort*, *microscope to support*, *dissolve to decide*. Report the identification with the tests used — "pile identified as nylon (burning, microscopy, formic-acid solubility)" — because a warranty dispute will ask exactly that question.

Glossary

Pile height — length of pile above the backing. **Tuft density** — tufts per unit area (rows × stitches). **Pile mass** — mass of pile yarn per unit area, excluding backing. **Tuft bind** — force to withdraw one tuft from the backing (ISO 4919). **Dynamic loading** — repeated impact simulating foot traffic. **Appearance retention** — graded visual change after simulated wear (hexapod, castor-chair). **Primary / secondary backing** — the sheet tufts pass through, and the stabilising sheet laminated beneath it. **Trilobal** — three-lobed fibre cross-section used to reduce visible soiling.

Key points

• Pile mass, tuft density and pile height are three different quantities — value and wear life follow pile mass and density, not height alone.
• Thickness (ISO 1765 principle) must be quoted at its measuring pressure; thickness loss after dynamic loading ranks flattening resistance.
• Recovery matters as much as loss: wool and nylon spring back, polypropylene flattens permanently.
• Tuft bind (ISO 4919) is reported as mean and minimum — failure starts at the weakest tuft.
• Fibre identification: burn to sort, microscope to support, dissolve to decide — never certify on the burning test alone.

Practice

1. A carpet has 10 rows/inch and 8 stitches/inch. Find the tuft density in tufts/in^2 and tufts/m^2 (1 in^2 = 6.45 cm^2; 1 m^2 = 10,000 cm^2).
2. Total carpet mass 2,850 g/m^2, backing system 1,050 g/m^2. Find the pile mass and say whether it exceeds a 1,600 g/m^2 minimum.
3. Thickness falls from 11.2 mm to 9.4 mm after dynamic loading. Calculate the percentage thickness loss and judge it against a 20 % maximum.
4. An unknown pile fibre curls from the flame, burns slowly with a burning-hair smell, self-extinguishes leaving a crushable char, shows scales under the microscope and dissolves in hot dilute alkali. Identify the fibre, naming the evidence from each test.

Answer key

1. Tuft density = 10 × 8 = **80 tufts/in^2**. Per cm^2: 80 / 6.45 = 12.4 tufts/cm^2; per m^2: 12.4 × 10,000 = about **124,000 tufts/m^2**.
2. Pile mass = 2,850 − 1,050 = **1,800 g/m^2** — exceeds the 1,600 g/m^2 minimum by 200 g/m^2 (pass, with margin).
3. Loss = 11.2 − 9.4 = 1.8 mm; % loss = 1.8 / 11.2 × 100 = **16.1 %** — within the 20 % maximum (pass). Recovery after rest should still be checked before final acceptance.
4. The fibre is **wool**. Burning: curling away, slow burning, self-extinguishing and burning-hair odour are characteristic of a protein fibre. Residue: crushable char bead, not a hard melt bead, excludes the synthetics. Microscopy: surface scales confirm wool (or a hair fibre). Solubility: dissolution in hot dilute alkali confirms a protein fibre and excludes nylon, polypropylene and acrylic.